Phase 2: Research Gap and Extension
Is the method ranking in M4 robust to the masking baseline?

Course: DA351 Explainable AI **Team / members / roll numbers:** [fill in]

Colour key: yellow = PUBLISHED results from the original paper (not our experiments) green = results from OUR experiments (this repository)

1. Original paper

Xuhong Li, Mengnan Du, Jiamin Chen, Yekun Chai, Himabindu Lakkaraju, Haoyi Xiong. *M^4: A Unified XAI Benchmark for Faithfulness Evaluation of Feature Attribution Methods across Metrics, Modalities and Models.* NeurIPS 2023, Datasets and Benchmarks Track. (Full PDF submitted separately.)

2. Research gap

M4 scores feature-attribution methods with perturbation metrics (MoRF, LeRF, ABPC) that replace the top-k or bottom-k features by a baseline value. The paper's definition (Eq. 1-3, page 4, below) says only that features are "masked"; the main text does not state the masking value or test other values. In the authors' pipeline, as ported in `m4bench/common.py`, the value is the constant grey (127,127,127) for images. A model's response to such inputs mixes attribution quality with how the model reacts to unusual inputs, so rankings could depend on this choice. **Gap:** to our knowledge, nobody has checked whether M4's method rankings survive a change of masking baseline. Related work shows the issue for deletion/insertion metrics in general (Picard et al. 2025, Rong et al. 2022, Hooker et al. 2019, Hsieh et al. 2021, Gomez et al. 2022), but not for M4's benchmark.

3. Published results (original paper)

PUBLISHED, M4 paper, page 4: Equations (1)-(3), the metric definitions

features will noticeably degrade the predicted probability of a model, while irrelevant features will have minimal impact on the probability mass. Several widely utilized metrics are built upon this fundamental idea. For example, Samek et al. [42] proposed *ordered perturbation*, which perturbs the input features gradually following the same order of values from the explanation results. They then calculate the area under the perturbation curve. The MoRF (most relevant first) metric represents the descending order, the LeRF (least relevant first) denotes the ascending order, and the ABPC (area between the perturbation curves) signifies their difference. Similarly deletion and insertion metrics [38] are an equivalent pair of MoRF and LeRF. Another example is *random perturbation*, such as Infidelity [53], which randomly perturbs the input, unlike ordered perturbation, and computes the empirical average of Eq.(4). Formally, their formulations are given as follows:

$$\mathrm{MoRF}(\boldsymbol{x}) = \frac{1}{L+1}\sum_{k=0}^{L}(\boldsymbol{f}(\boldsymbol{x}^{(0)}_{\mathrm{MoRF}}) - \boldsymbol{f}(\boldsymbol{x}^{(k)}_{\mathrm{MoRF}}))\,, \tag{1}$$

$$\mathrm{LeRF}(\boldsymbol{x}) = \frac{1}{L+1}\sum_{k=0}^{L}(\boldsymbol{f}(\boldsymbol{x}^{(0)}_{\mathrm{LeRF}}) - \boldsymbol{f}(\boldsymbol{x}^{(k)}_{\mathrm{LeRF}}))\,, \tag{2}$$

$$\mathrm{ABPC}(\boldsymbol{x}) = \frac{1}{L+1}\sum_{k=0}^{L}(\boldsymbol{f}(\boldsymbol{x}^{(k)}_{\mathrm{LeRF}}) - \boldsymbol{f}(\boldsymbol{x}^{(k)}_{\mathrm{MoRF}}))\,, \tag{3}$$

$$\mathrm{INFD}(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{I}\sim\mu_I}(\boldsymbol{I}^T\mathcal{A}(\boldsymbol{x},\boldsymbol{f}) - (\boldsymbol{f}(\boldsymbol{x}) - \boldsymbol{f}(\boldsymbol{x}-\boldsymbol{I}))^2), \tag{4}$$

where $\boldsymbol{f}$ is the DNN model including the architecture and the trained parameters, $\boldsymbol{x}^{(0)}$ is the original input, $\boldsymbol{x}^{(k)}_{\mathrm{MoRF}}$ is the perturbed input whose top-k features are masked, $\boldsymbol{x}^{(k)}_{\mathrm{LeRF}}$ is the perturbed input whose bottom-k features are masked, and $\mathcal{A}$ is a feature attribution method taking a data sample $\boldsymbol{x}$ and a trained model $\boldsymbol{f}$ as input. Note that if the explanation is of better quality and of better loyalty to the

PUBLISHED, M4 paper, page 6: Table 1, ResNet-50

Table 1: Evaluation results on the ImageNet-pretrained ResNet-50, with constant and random baselines, GradCAM, IG and SG. "Random-16" indicates that the random values are given at patch level (16×16 pixels) while "Random" is at pixel level.

Attribution Methods	MoRF ↑	ABPC ↑	PScore ↑	INFD ↓	SynScore ↑
Constant	0.000	0.000	N/A	3.015	0.072
Random-16	0.596	0.007	N/A	3.039	0.077
Random	0.599	0.008	N/A	3.015	0.078
GradCAM	0.628	**0.424**	**0.835**	2.496	**1.000**
IG	**0.709**	0.377	0.812	2.373	0.999
SG	0.701	0.369	0.820	**2.323**	0.998

4. Original approach, reproduced (our run, grey baseline)

Our PyTorch port of the M4 pipeline (`m4bench/`), ResNet-50, the first 500 images of M4's image list, single grey baseline. Published numbers are copied from Table 1, p. 6; the "ours" columns are our measurements.

method	MoRF paper (p.6)	MoRF ours	ABPC paper (p.6)	ABPC ours [95% CI]
GradCAM	0.628	0.657	0.424	0.455 [0.434, 0.475]
IG	0.709	0.739	0.377	0.466 [0.443, 0.488]
SG	0.701	0.728	0.369	0.432 [0.410, 0.455]

Observation. Our MoRF and ABPC are somewhat higher than the paper's. The paper's ABPC order is GradCAM > IG > SG, while ours under grey is IG > GradCAM > SG. The IG and GradCAM confidence intervals overlap, so this particular difference is not reliable. We did not investigate the source of the offset (candidates: image subset, weights conversion, implementation details).

5. Proposed improvement

Instead of reporting one ABPC under one baseline, evaluate every saved attribution map under **five masking baselines** and report (i) 95% bootstrap confidence intervals, (ii) Kendall's tau and the flipped method pairs against the grey ranking, (iii) a **multi-baseline ABPC** (mean over baselines) and (iv) the Baseline Sensitivity Score, BSS = standard deviation over baselines of the mean ABPC. The attributions are **not recomputed**: the baseline is the only thing that changes.

	M4 (before)	Proposed (after)
Masking baseline	one fixed grey value	grey, black, ImageNet-mean colour, Gaussian blur (sigma 11), uniform noise
Uncertainty	none reported	95% bootstrap CI over images
Ranking robustness	not tested	Kendall tau, flipped pairs, bootstrap reproducibility
Sample size	one fixed N	subset-vs-full ranking agreement (MobileNet, 5000 images)

Supporting references. Rong et al. (ICML 2022): pixel-perturbation evaluations give conflicting rankings, and protocols can be made more robust. Picard et al. (arXiv 2025, preprint): the baseline in deletion/insertion systematically favours some attribution methods. Hsieh et al. (ICLR 2021) and Hooker et al. (NeurIPS 2019): removal-based evaluation can be biased by the replacement value / distribution shift.

6. Implementation

- `rescore_baselines.py` (new): loads the saved M4 maps (`maps_sq.npy`), builds the MoRF/LeRF stacks with `baseline_image()` + `perturbation_stacks()`, which keep M4's 100-step percentile schedule and class of interest and replace only the fill value, and stores the class-probability curves.
- `analyze_baselines.py` (new): bootstrap CIs, Kendall tau, flipped pairs, multi-baseline ABPC, BSS, bootstrap ranking reproducibility, sample-size analysis, figures.
- The extension only adds these two files. It reuses the existing `m4bench/` image loading, model loading and metric helpers (`vision_metrics.batched_outputs`, `resize_map`) unchanged.
- **Same conditions:** same images, same saved maps, same schedule, fp32 forward passes. Check done during development on 20 images: re-scoring with grey reproduced the stored M4 curves (per-image ABPC correlation 0.9995; mean ABPC 0.5438 vs 0.5429).

7. Results after the improvement

7.1 Summary: does the grey (M4) method ranking survive other masking baselines?

model	images	ranking under grey (M4)	baselines that change the ranking	min Kendall tau
ResNet-50	500	ig>gradcam>sg	black, blur, noise	-0.333
ResNet-101	500	ig>gradcam>sg	black, blur, noise	-0.333

model	images	ranking under grey (M4)	baselines that change the ranking	min Kendall tau
ResNet-152	500	gradcam>ig>sg	black, mean, blur	-1.000
VGG16	500	ig>sg>gradcam	black	0.333
MobileNet	500	ig>sg>gradcam	black, mean, blur, noise	-1.000
ViT-S	100	ga>bt_head>bt_token>ig>sg	black, mean, blur	0.400
ViT-B	100	ig>bt_token>bt_head>sg>ga	black, blur, noise	-0.200
MAE-ViT-B	100	bt_head>bt_token>ga>ig>sg	black, noise	0.800

7.2 ABPC by masking baseline, mean [95% CI] (higher = more faithful)

ResNet-50, 500 images:

method	grey	black	mean	blur	noise
gradcam	0.455 [0.434, 0.475]	0.451 [0.430, 0.471]	0.456 [0.436, 0.475]	0.456 [0.436, 0.475]	0.433 [0.412, 0.453]
ig	0.466 [0.443, 0.488]	0.287 [0.265, 0.310]	0.471 [0.449, 0.493]	0.481 [0.458, 0.501]	0.389 [0.367, 0.411]
sg	0.432 [0.410, 0.455]	0.308 [0.286, 0.329]	0.435 [0.412, 0.457]	0.489 [0.467, 0.509]	0.367 [0.345, 0.390]

ResNet-101, 500 images:

method	grey	black	mean	blur	noise
gradcam	0.475 [0.454, 0.497]	0.464 [0.442, 0.485]	0.478 [0.457, 0.499]	0.493 [0.471, 0.513]	0.461 [0.440, 0.483]
ig	0.479 [0.456, 0.503]	0.284 [0.263, 0.304]	0.484 [0.461, 0.507]	0.498 [0.476, 0.520]	0.389 [0.368, 0.412]
sg	0.446 [0.424, 0.469]	0.303 [0.282, 0.325]	0.453 [0.431, 0.476]	0.510 [0.489, 0.531]	0.371 [0.349, 0.393]

ResNet-152, 500 images:

method	grey	black	mean	blur	noise
gradcam	0.493 [0.472, 0.513]	0.486 [0.465, 0.507]	0.494 [0.473, 0.514]	0.486 [0.465, 0.505]	0.482 [0.462, 0.502]
ig	0.490 [0.466,	0.272 [0.253,	0.496 [0.473,	0.497 [0.475,	0.377 [0.355,

method	grey	black	mean	blur	noise
	0.513]	0.293]	0.519]	0.520]	0.400]
sg	0.446 [0.424, 0.468]	0.291 [0.270, 0.311]	0.454 [0.430, 0.477]	0.504 [0.482, 0.526]	0.368 [0.347, 0.390]

VGG16, 500 images:

method	grey	black	mean	blur	noise
gradcam	0.196 [0.165, 0.228]	0.177 [0.148, 0.210]	0.196 [0.164, 0.226]	0.221 [0.188, 0.254]	0.173 [0.143, 0.203]
ig	0.475 [0.451, 0.498]	0.262 [0.239, 0.286]	0.482 [0.458, 0.506]	0.510 [0.487, 0.533]	0.395 [0.369, 0.420]
sg	0.423 [0.398, 0.448]	0.269 [0.247, 0.291]	0.432 [0.409, 0.456]	0.507 [0.483, 0.529]	0.363 [0.338, 0.389]

MobileNet, 500 images:

method	grey	black	mean	blur	noise
gradcam	0.330 [0.309, 0.354]	0.339 [0.316, 0.362]	0.349 [0.326, 0.373]	0.342 [0.318, 0.364]	0.320 [0.296, 0.343]
ig	0.371 [0.349, 0.392]	0.186 [0.170, 0.202]	0.378 [0.356, 0.401]	0.375 [0.354, 0.396]	0.293 [0.272, 0.313]
sg	0.339 [0.318, 0.360]	0.225 [0.208, 0.243]	0.341 [0.320, 0.362]	0.396 [0.378, 0.417]	0.295 [0.277, 0.314]

ViT-S, 100 images:

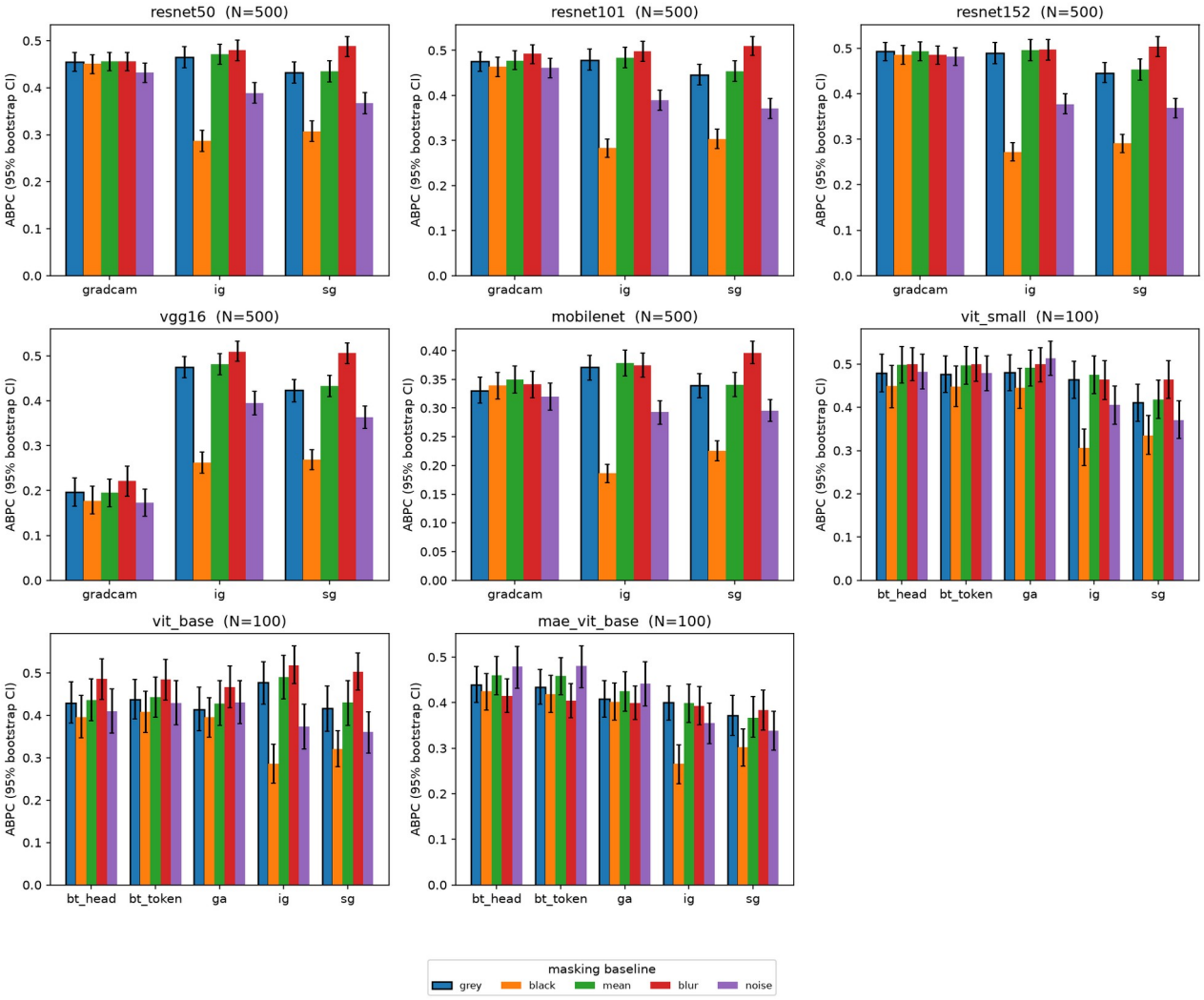
method	grey	black	mean	blur	noise
bt_head	0.479 [0.435, 0.522]	0.449 [0.399, 0.497]	0.498 [0.456, 0.540]	0.500 [0.461, 0.538]	0.482 [0.442, 0.523]
bt_token	0.477 [0.434, 0.519]	0.448 [0.402, 0.496]	0.497 [0.453, 0.541]	0.499 [0.461, 0.538]	0.479 [0.439, 0.519]
ga	0.481 [0.438, 0.521]	0.445 [0.398, 0.490]	0.491 [0.449, 0.532]	0.500 [0.458, 0.538]	0.513 [0.474, 0.553]
ig	0.464 [0.421, 0.507]	0.307 [0.266, 0.350]	0.475 [0.431, 0.518]	0.464 [0.419, 0.508]	0.406 [0.361, 0.449]
sg	0.410 [0.367, 0.454]	0.335 [0.292, 0.382]	0.418 [0.374, 0.463]	0.465 [0.420, 0.508]	0.370 [0.329, 0.415]

ViT-B, 100 images:

method	grey	black	mean	blur	noise
bt_head	0.430 [0.382, 0.479]	0.395 [0.347, 0.447]	0.435 [0.387, 0.486]	0.486 [0.437, 0.533]	0.410 [0.358, 0.462]
bt_token	0.438 [0.392, 0.485]	0.408 [0.360, 0.457]	0.444 [0.395, 0.491]	0.485 [0.436, 0.532]	0.429 [0.378, 0.483]
ga	0.414 [0.364, 0.466]	0.396 [0.348, 0.442]	0.427 [0.376, 0.481]	0.467 [0.418, 0.517]	0.430 [0.381, 0.482]
ig	0.477 [0.427, 0.526]	0.286 [0.241, 0.331]	0.490 [0.438, 0.541]	0.518 [0.474, 0.565]	0.373 [0.320, 0.426]
sg	0.417 [0.362, 0.470]	0.320 [0.279, 0.363]	0.430 [0.377, 0.481]	0.503 [0.459, 0.547]	0.361 [0.310, 0.409]

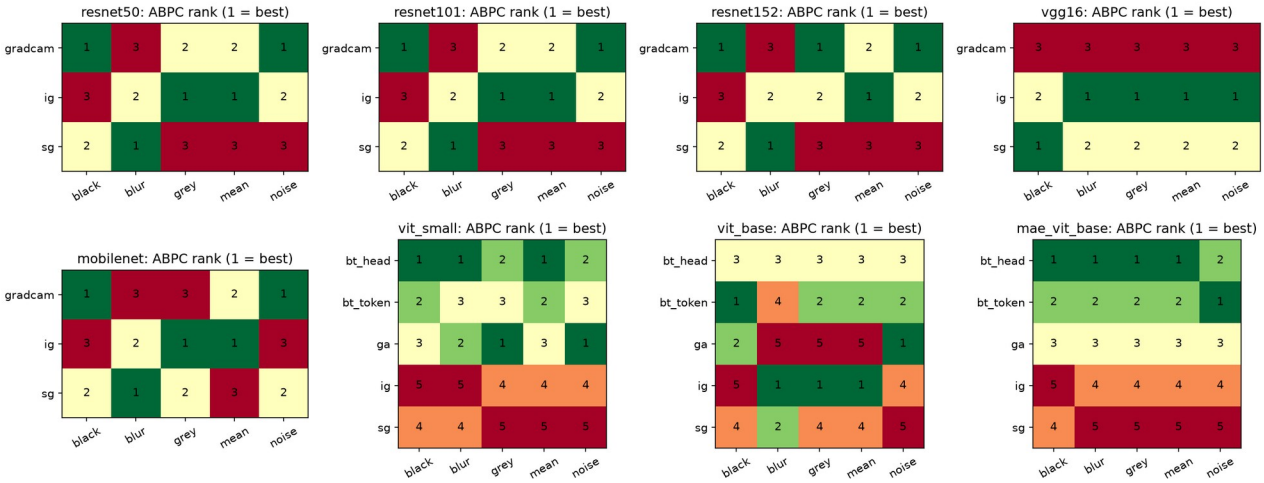
MAE-ViT-B, 100 images:

method	grey	black	mean	blur	noise
bt_head	0.439 [0.401, 0.479]	0.425 [0.383, 0.464]	0.460 [0.417, 0.501]	0.414 [0.379, 0.452]	0.479 [0.432, 0.523]
bt_token	0.434 [0.397, 0.473]	0.419 [0.379, 0.460]	0.459 [0.418, 0.498]	0.404 [0.367, 0.441]	0.481 [0.433, 0.525]
ga	0.408 [0.368, 0.448]	0.402 [0.362, 0.443]	0.425 [0.381, 0.468]	0.400 [0.363, 0.437]	0.442 [0.393, 0.489]
ig	0.401 [0.361, 0.437]	0.266 [0.222, 0.308]	0.399 [0.356, 0.440]	0.393 [0.352, 0.435]	0.355 [0.310, 0.399]
sg	0.372 [0.328, 0.416]	0.302 [0.260, 0.342]	0.367 [0.324, 0.413]	0.383 [0.339, 0.428]	0.339 [0.296, 0.382]



7.3 Ranking under every baseline (Kendall tau and flipped pairs vs grey)

Per-model Kendall tau and the exact method pairs that flip under each baseline are in `phase2_results/rescore/summary/rank_stability.csv`. Heatmap: ABPC rank of each method under each baseline (a rank that changes along a row means the ranking is baseline-dependent).



7.4 Before (grey) vs after (multi-baseline)

model	method	ABPC grey (before)	ABPC multi-baseline (after)	BSS	rank before	rank after
ResNet-50	gradcam	0.455	0.450	0.009	2	1

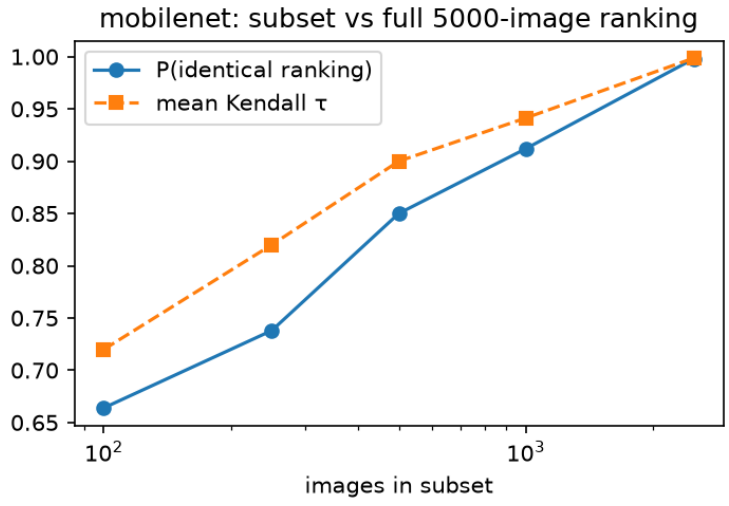
model	method	ABPC grey (before)	ABPC multi-baseline (after)	BSS	rank before	rank after
ResNet-50	ig	0.466	0.419	0.074	1	2
ResNet-50	sg	0.432	0.406	0.062	3	3
ResNet-101	gradcam	0.475	0.474	0.011	2	1
ResNet-101	ig	0.479	0.427	0.081	1	2
ResNet-101	sg	0.446	0.417	0.072	3	3
ResNet-152	gradcam	0.493	0.488	0.005	1	1
ResNet-152	ig	0.490	0.426	0.090	2	2
ResNet-152	sg	0.446	0.413	0.075	3	3
VGG16	gradcam	0.196	0.193	0.017	3	3
VGG16	ig	0.475	0.425	0.090	1	1
VGG16	sg	0.423	0.399	0.080	2	2
MobileNet	gradcam	0.330	0.336	0.010	3	1
MobileNet	ig	0.371	0.321	0.074	1	2
MobileNet	sg	0.339	0.319	0.057	2	3
ViT-S	bt_head	0.479	0.482	0.018	2	2
ViT-S	bt_token	0.477	0.480	0.018	3	3
ViT-S	ga	0.481	0.486	0.023	1	1
ViT-S	ig	0.464	0.423	0.063	4	4
ViT-S	sg	0.410	0.400	0.044	5	5
ViT-B	bt_head	0.430	0.431	0.031	3	2
ViT-B	bt_token	0.438	0.441	0.025	2	1
ViT-B	ga	0.414	0.427	0.023	5	4
ViT-B	ig	0.477	0.429	0.087	1	3

model	method	ABPC grey (before)	ABPC multi-baseline (after)	BSS	rank before	rank after
ViT-B	sg	0.417	0.406	0.062	4	5
MAE-ViT-B	bt_head	0.439	0.444	0.023	1	1
MAE-ViT-B	bt_token	0.434	0.439	0.027	2	2
MAE-ViT-B	ga	0.408	0.415	0.016	3	3
MAE-ViT-B	ig	0.401	0.363	0.051	4	4
MAE-ViT-B	sg	0.372	0.353	0.029	5	5

ResNet-50: bootstrap reproducibility of the full ranking, grey 84% vs multi-baseline 100%. ResNet-101: bootstrap reproducibility of the full ranking, grey 62% vs multi-baseline 100%. ResNet-152: bootstrap reproducibility of the full ranking, grey 66% vs multi-baseline 100%. VGG16: bootstrap reproducibility of the full ranking, grey 100% vs multi-baseline 100%. MobileNet: bootstrap reproducibility of the full ranking, grey 74% vs multi-baseline 58%. ViT-S: bootstrap reproducibility of the full ranking, grey 32% vs multi-baseline 59%. ViT-B: bootstrap reproducibility of the full ranking, grey 38% vs multi-baseline 18%. MAE-ViT-B: bootstrap reproducibility of the full ranking, grey 57% vs multi-baseline 69%.

7.5 How much does a 500-image subset matter? (MobileNet, grey baseline, M4's own per-image scores on all 5000 images)

images in subset	P(same ranking as full 5000)	mean Kendall tau
100	0.664	0.720
250	0.738	0.820
500	0.850	0.900
1000	0.912	0.941
2500	0.998	0.999



8. Interpretation

- **The grey ranking is not stable across baselines.** It changes under at least one other baseline in 8 of 8 models (ResNet-50: black, blur, noise; ResNet-101: black, blur, noise; ResNet-152: black, mean, blur; VGG16: black; MobileNet: black, mean, blur, noise; ViT-S: black, mean, blur; ViT-B: black, blur, noise; MAE-ViT-B: black, noise). A complete reversal of the ranking occurs for ResNet-152 under blur; MobileNet under black; MobileNet under noise. For ResNet-50 specifically, M4's order (ig > gradcam > sg) becomes gradcam > sg > ig under black, gradcam > ig > sg under noise and sg > ig > gradcam under blur.
- **IG and SG are the baseline-sensitive methods.** Their BSS ranges from 0.029 to 0.090 across models, versus 0.005 to 0.031 for GradCAM and the attention-based methods; the pixel-level methods have the higher BSS in 8 of 8 models (ResNet-50, ResNet-101, ResNet-152, VGG16, MobileNet, ViT-S, ViT-B, MAE-ViT-B). Under black, IG or SG fall below every coarse method in 14 of 16 (model, method) cases, and IG loses on average 0.18 ABPC going from grey to black (range 0.13 to 0.22). *Why* (scattered pixel-level masks vs smooth masks) is a hypothesis we did not test.
- **Methods that cannot be separated.** For ViT-S, ViT-B, MAE-ViT-B, all coarse-method grey-baseline confidence intervals overlap, so their order, and its changes across baselines, are not reliable. Images per model: ViT-S 100, ViT-B 100, MAE-ViT-B 100.
- **Multi-baseline ranking.** ResNet-50: ig > gradcam > sg (grey, reproduced in 84% of bootstrap resamples) vs gradcam > ig > sg (100%); ResNet-101: ig > gradcam > sg (grey, reproduced in 62% of bootstrap resamples) vs gradcam > ig > sg (100%); ResNet-152: gradcam > ig > sg (grey, reproduced in 66% of bootstrap resamples) vs gradcam > ig > sg (100%); VGG16: ig > sg > gradcam (grey, reproduced in 100% of bootstrap resamples) vs ig > sg > gradcam (100%); MobileNet: ig > sg > gradcam (grey, reproduced in 74% of bootstrap resamples) vs gradcam > ig > sg (58%); ViT-S: ga > bt_head > bt_token > ig > sg (grey, reproduced in 32% of bootstrap resamples) vs ga > bt_head > bt_token > ig > sg (59%); ViT-B: ig > bt_token > bt_head > sg > ga (grey, reproduced in 38% of bootstrap resamples) vs bt_token > bt_head > ig > ga > sg (18%); MAE-ViT-B: bt_head > bt_token > ga > ig > sg (grey, reproduced in 57% of bootstrap resamples) vs bt_head > bt_token > ga > ig > sg (69%). This means *more or less reproducible*, not *more correct*: the result depends on which baselines are averaged. Grey (127) and the ImageNet-mean colour are close in value, yet even this small change alters the ranking in 3 of 8 models (ResNet-152, MobileNet, ViT-S).
- **Sample size:** a random 500-image subset reproduces the full 5000-image MobileNet ranking in only 85% of draws, so the 500-image runs carry real uncertainty (hence the confidence intervals). In our own 500-image MobileNet rescoring the grey ranking is ig > sg > gradcam, whereas M4's full 5000-image ranking is gradcam > ig > sg, so even the grey reference for that model is a subset artefact.

Does the improvement address the gap? Yes for the scope tested (8 vision models): it shows how M4-style rankings depend on the masking baseline and quantifies how much, which the single-baseline protocol cannot show. It does not show that any one baseline is the "right" one.

9. Limitations

- 8 of M4's 9 vision models (ViT-L not run) and small image subsets (500 for the CNNs, 100 for the ViTs). No text models: [PAD]/[MASK]/[UNK] were not run, and no PScore/SynScore under other baselines.
- The CNN runs use 500 images and the ViT runs 100, so comparisons *between* architectures are weaker than comparisons *within* a model.
- The saved attribution maps use the squared-channel reduction; M4's own metric uses the absolute reduction. Grey re-scoring still reproduced the stored curves closely (see section 6).
- Our replication differs from the paper's Table 1 numbers (section 4); the cause was not investigated.
- Cited papers were checked at abstract / search-excerpt level only.

10. References

- Li, X., Du, M., Chen, J., Chai, Y., Lakkaraju, H., Xiong, H. (2023). M4: A unified XAI benchmark for faithfulness evaluation of feature attribution methods across metrics, modalities and models. NeurIPS (Datasets and Benchmarks).
- Rong, Y., Leemann, T., Borisov, V., Kasneci, G., Kasneci, E. (2022). A consistent and efficient evaluation strategy for attribution methods. ICML, PMLR 162:18770-18795.
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- Hsieh, C.-Y., Yeh, C.-K., Liu, X., Ravikumar, P., Kim, S., Kumar, S., Hsieh, C.-J. (2021). Evaluations and methods for explanation through robustness analysis. ICLR.
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networks. NeurIPS.
- Gomez, T., Freour, T., Mouchere, H. (2022). Metrics for saliency map evaluation of deep learning explanation methods. arXiv:2201.13291 (preprint).
- Petsiuk, V., Das, A., Saenko, K. (2018). RISE: Randomized input sampling for explanation of black-box models. arXiv:1806.07421.

Provenance

Published values: only Table 1 (p. 6) and Equations 1-3 (p. 4), shown as page images and in the "paper" columns of section 4. Everything else comes from the scripts in this repository run on the saved M4 attribution maps; raw curves and logs are in `phase2_results/`; regenerate with the commands in `README.md`.